

Miao Lu

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EDUCATION

Stanford University

Ph.D. Candidate in Operations Research. Advisor: Prof. Jose Blanchet

Thesis: "Efficient and Robust Reinforcement Learning: from Theory to Self-evolving Generative AI"

Stanford, USA

Sep.2023 – Dec.2026 (expected)

University of Science and Technology of China

B.S. in Mathematics & Applied Mathematics. Ranking: 2/140, Guo Moruo Scholarship (summa cum laude)

Hefei, China

Sep.2018 – Jun.2022

RESEARCH HIGHLIGHT

I study reinforcement learning and LLM post-training, with a focus on long-horizon and self-evolving agents. My research also investigates the theoretical foundations of sample efficiency in RL.

Thread 1: Post-training for Long-horizon & Self-evolving Agents

I develop post-training methods for LLM agents to actively manage long contexts and improve through self-generated data.

Through end-to-end RL, **SUPO** [C11] jointly trains context compression and tool use, while **Context-Folding** [C13] learns to branch into subtasks and fold their results back into the main thread. **INFUSER** [C16] co-trains a document-grounded question generator and a solver, using optimizer-aware influence scores to elicit questions that benefit the solver and adapt its curriculum.

[C11] Beyond the Context Window: Scaling Agentic RL via End-to-end Optimized Context Compression. *ACL 2026*.

[C13] Scaling Long-Horizon LLM Agent via Context Folding. *ICML 2026*.

[C16] INFUSER: Influence-Guided Self-Evolution Improves Reasoning. *NeurIPS 2026*.

Thread 2: Algorithmic & Theoretical Foundations of RL

I design RL algorithms and theory that address data uncertainty and guide exploration.

RPO [C9] (Regularized Preference Optimization) addresses reward uncertainty in RLHF through an adversarial formulation, with a mathematically equivalent objective combining DPO and SFT objectives, mitigating overoptimization with sample-efficiency guarantees under partial coverage data assumption. **MEX** [C5] (Maximize to Explore) unifies estimation, planning, and exploration in one objective, with sublinear-regret guarantees under general function approximations.

[C9] Provably Mitigating Overoptimization in RLHF: Your SFT Loss is Implicitly an Adversarial Regularizer. *NeurIPS 2024*.

[C5] Maximize to Explore: One Objective Function Fusing Estimation, Planning, and Exploration. *NeurIPS 2023 Spotlight*.

Thread 3: Optimization in Deep Learning

I investigate how deep learning optimization landscape shapes feature learning and optimization dynamics.

Benign Oscillation [C7] shows, in a feature-noise model, how large-learning-rate SGD can learn weak features missed at smaller learning rates. **Momentum in River-Valley Landscapes** [C14] explains how heavy-ball momentum stabilizes large-learning-rate oscillations, accelerating progress along the low-loss river in a river-valley loss landscape.

[C7] Benign Oscillation of Stochastic Gradient Descent with Large Learning Rates. *ICLR 2024*.

[C14] Towards Understanding Momentum Acceleration in River-Valley Loss Landscape. *NeurIPS 2026 / ICML HiLD 2026 Spotlight*.

INDUSTRIAL EXPERIENCES

NVIDIA Research

Research Scientist Intern

Research on LLM long-context training data synthesis and continued pretraining [P3].

Santa Clara, USA

Jan.2026 – Jul.2026

ByteDance Seed

Research Scientist Intern

Research on RL algorithms to scale LLM agent training for long horizon coding/deep research tasks beyond a fixed context limit, with context management techniques of context compression and context-folding [C11][C13].

San Jose, USA

Jun.2025 – Jan.2026

Citadel Securities

Quantitative Research Intern

Systematic Equities Team, AI for alpha research.

Miami, USA

Jul.2026 – Sep.2026

Ubiquant Investment

Quantitative Research Intern

Ubiquant AI Lab, research on DL and RL for quantitative trading.

Shanghai, China

Jul.2022 – Sep.2022

Authors with * contributed equally to the work, and † represents alphabetical order.

- [P5] Zhiwei Xu, Jiaqi Pan, Zhihan Liu, David D. Baek, **Miao Lu**, Yanqing Liu, Buhuang Liu, Yixin Wang, Hrayr Harutyunyan, Yanhan Hou, Haichao Yu. Turn-Level Critic Guidance for Training and Inference in Terminal Agents. *Preprint under conference review*.
- [P4] Junze Ye, **Miao Lu**, Jose Blanchet, Mohsen Bayati. Value-Filtered Expert Query for Efficient Language Agent Finetuning: an Empirical Study. *NeurIPS workshop on Mathematical Foundations and Operational Integration of Machine Learning for Uncertainty-Aware Decision-Making (MLxOR)* 2026.
- [P3] **Miao Lu**, Zhiding Yu, Jose Blanchet, Oreste Villa, Yaosheng Fu. LDIS: Efficient Long-context Training Data Synthesis via Interleaved-Study. *NeurIPS workshop on Long Context Foundation Models (LCFM)* 2026.
- [P2] **Miao Lu**, Yuxuan Han, Han Zhong, Zhengyuan Zhou, Jose Blanchet. Robust Assortment Optimization from Observational Data. *arXiv preprint, under journal submission*.
- [P1] Yuxuan Han, Han Zhong, **Miao Lu**, Jose Blanchet, Zhengyuan Zhou. Learning an Optimal Assortment Policy under Observational Data. *arXiv preprint, under review at Management Science (MS)*.
- [C16] Siyu Chen, **Miao Lu**, Beining Wu, Heejune Sheen, Fengzhuo Zhang, Shuangning Li, Zhiyuan Li, Jose Blanchet, Tianhao Wang, Zhuoran Yang. INFUSER: Influence-Guided Self-Evolution Improves Reasoning. *Neural Information Processing Systems (NeurIPS) 2026; COLM Workshop on Scientific Understanding of Foundation Models (SciFM)* 2026.
- [C15] Junze Ye*, Jiayi Cheng*, **Miao Lu***, Michal Mankowski, Jose Blanchet, Mohsen Bayati. A Few Teacher Steps Go a Long Way: Cost-Efficient On-Policy Data Augmentation for Agent Post-Training. *Neural Information Processing Systems (NeurIPS) 2026; ICML Workshop on RL from World Feedback (RLxF)* 2026.
- [C14] **Miao Lu**, Zeyu Bian, Kaiyue Wen, Beining Wu, Siyu Chen, Tianhao Wang, Zhiyuan Li. Towards Understanding Momentum Acceleration in River-Valley Loss Landscape. *Neural Information Processing Systems (NeurIPS) 2026; ICML Workshop on High-dimensional Learning Dynamics (HiLD)* 2026 **Spotlight**.
- [C13] Weiwei Sun, **Miao Lu**, Zhan Ling, Xuesong Yao, Kang Liu, Yiming Yang, Jiecao Chen. Scaling Long-Horizon LLM Agent via Context Folding. *International Conference on Machine Learning (ICML)* 2026.
- [C12] Xingwu Chen*, **Miao Lu***, Beining Wu, Difan Zou. Towards Theoretical Understanding of Transformer Test-Time Computing: Investigation on In-Context Linear Regression. *NeurIPS Workshop on Foundations of Reasoning in Language Models (FoRLM)* 2025; *International Conference on Machine Learning (ICML)* 2026.
- [C11] **Miao Lu**, Weiwei Sun, Weihua Du, Zhan Ling, Xuesong Yao, Kang Liu, Jiecao Chen. Beyond the Context Window: Scaling Agentic RL via End-to-end Optimized Context Compression. *Association for Computational Linguistics (ACL)* 2026.
- [C10] Siyu Chen*, Beining Wu*, **Miao Lu**, Zhuoran Yang, Tianhao Wang. Can Neural Networks Achieve Optimal Computational-statistical Tradeoff? An Analysis on Single-Index Model. *International Conference on Learning Representations (ICLR) 2025* **Oral presentation**; *NeurIPS Workshop on Mathematics of Modern Machine Learning (M3L)* 2024 **Oral presentation**.
- [C9] Zhihan Liu*, **Miao Lu***, Shenao Zhang, Boyi Liu, Hongyi Guo, Yingxiang Yang, Jose Blanchet, Zhaoran Wang. Provably Mitigating Overoptimization in RLHF: Your SFT Loss is Implicitly an Adversarial Regularizer. *Neural Information Processing Systems (NeurIPS) 2024; ICML Workshop on Aligning Reinforcement Learning Experimentalists and Theorists (ARLET)* 2024.
- [C8] **Miao Lu***, Han Zhong*, Tong Zhang, Jose Blanchet. Distributionally Robust Reinforcement Learning with Interactive Data Collection: Fundamental Hardness and Near-Optimal Algorithm. *Neural Information Processing Systems (NeurIPS) 2024; ICML Workshop on Aligning Reinforcement Learning Experimentalists and Theorists (ARLET)* 2024.
- [C7] **Miao Lu***, Beining Wu*, Xiaodong Yang, Difan Zou. Benign Oscillation of Stochastic Gradient Descent with Large Learning Rates. *International Conference on Learning Representations (ICLR) 2024; NeurIPS Workshop on Mathematics of Modern Machine Learning (M3L)* 2023.
- [C6] Jose Blanchet†, **Miao Lu†**, Tong Zhang†, Han Zhong†. Double Pessimism is Provably Efficient for Distributionally Robust Offline Reinforcement Learning: Generic Algorithm and Robust Partial Coverage. *Neural Information Processing Systems (NeurIPS) 2023; Extended version under major revision at Mathematics of Operations Research (MOR)*.
- [C5] Zhihan Liu*, **Miao Lu***, Wei Xiong*, Han Zhong, Hao Hu, Shenao Zhang, Sirui Zheng, Zhuoran Yang, Zhaoran Wang. Maximize to Explore: One Objective Function Fusing Estimation, Planning, and Exploration. *Neural Information Processing Systems (NeurIPS) 2023* **Spotlight**; *Extended version under review at Operations Research (OR)*.
- [C4] **Miao Lu**, Yifei Min, Zhaoran Wang, Zhuoran Yang. Pessimism in the Face of Confounders: Provably Efficient Offline Reinforcement Learning in Partially Observable Markov Decision Processes. *International Conference on Learning Representations (ICLR)* 2023.
- [C3] Zhihan Liu*, **Miao Lu***, Zhaoran Wang, Michael I. Jordan, Zhuoran Yang. Welfare Maximization in Competitive Equilibrium: Reinforcement Learning for Markov Exchange Economy. *International Conference on Machine Learning (ICML)* 2022.
- [C2] **Miao Lu***, Xiaolong Luo*, Tianlong Chen, Wuyang Chen, Dong Liu, Zhangyang Wang. Learning Pruning-Friendly Networks via Frank-Wolfe: One-Shot, Any-Sparsity, and No Retraining. *International Conference on Learning Representations (ICLR) 2022* **Spotlight**.
- [C1] Yufei Kuang, **Miao Lu**, Jie Wang, Qi Zhou, Bin Li, Houqiang Li. Learning Robust Policy against Disturbance in Transition Dynamics via State-Conservative Policy Optimization. *Association for Advancement of Artificial Intelligence (AAAI)* 2022.

INVITED TALKS

Scaling Long Horizon LLM Agent via Reinforcement Learning [C11][C13] o TikTok Research Seminar, remote [C11][C13]	Oct.2025
Theoretical Foundations of Distributionally Robust Reinforcement Learning [C6][C8][P2] o 2025 INFORMS annual meeting, Atlanta GA, USA [P1][P2] o 2024 INFORMS annual meeting, Seattle, WA, USA [C8] o 58th Annual Conference on Information Sciences and Systems (CISS), Princeton, NJ, USA [C6] o 2023 INFORMS annual meeting, Phoenix, AZ, USA [C6]	Oct.2025 Oct.2024 Mar.2024 Oct.2023
Computational-statistical Trade-off of Learning Single-index Models via Neural Networks [C10] o 2nd Mathematics of Modern Machine Learning Workshop (M3L), Vancouver, BC, Canada	Dec.2024

AWARDS AND HONORS

Xinhe Scholarship (outstanding undergraduate researchers, School of the Gifted Young, USTC)	Mar.2023
Yuanqing Yang Scholarship (top scholarship, School of Mathematical Sciences, USTC)	Jan.2022
The 41st Guo Moruo Scholarship (highest honor, USTC)	Dec.2021
Chinese National Scholarship (top scholarship, Ministry of Education of China)	Nov.2019, 2020

ACADEMIC SERVICES

Journal Reviewer: Annals of Applied Probability (AOAP), Operations Research (OR), Management Science (MS), Mathematics of Operations Research (MOR), Transactions on Machine Learning Research (TMLR)

Conference Reviewer: Neural Information Processing Systems (NeurIPS; 2023-2026), International Conference on Machine Learning (ICML; 2024-2026), International Conference on Learning Representations (ICLR; 2024-2026), International Conference on Artificial Intelligence and Statistics (AISTATS; 2025), ICML Workshop on Aligning Reinforcement Learning Experimentalists and Theorists (ARLET; 2024), ICML Workshop on Exploration in AI Today (EXAIT; 2025), ICML Workshop on Methods and Opportunities at Small Scale (MOSS; 2025), NeurIPS Workshop on Mathematics of Modern Machine Learning (M3L; 2024), Association for the Advancement of Artificial Intelligence (AAAI; 2025)